Tensor computations

Irreducibility of asymptotic tensor-rank sublevel varieties

Difficulty: extreme **Importance:** interesting to specialist

Status: Solved

Rating rationale: Controlling irreducibility for every tensor-power cost sublevel is a foundational geometric question stronger than finiteness of the attainable costs. Its immediate audience is specialists in the algebraic geometry of tensor complexity. These ratings are historical and refer to the original universal question, which is resolved negatively below. They do not rate the remaining order-three question.

Resolution — 3 November 2026

Resolved negatively. Wenqi Zhu (University of Oxford), Qiyuan Chen (Academy of Mathematics and Systems Science, Chinese Academy of Sciences) and Ping Nie (University of Waterloo).

The answer to the universal question is **no**. Over $\mathbb{C}$, in the 54-dimensional space $V = \mathbb{C}^3 \otimes \mathbb{C}^3 \otimes \mathbb{C}^3 \otimes \mathbb{C}^2$ of order four, the sublevel

$$V_{\le a} = \{T \in V : \widetilde{R}(T) \le a\}, \qquad a = \sqrt[3]{62},$$

is the union of two proper relatively Zariski-closed subsets, so it is not irreducible. This is Theorem 1.1, stated on p. 1 of the 16-page paper *Asymptotic Tensor-Rank Sublevels Need Not Be Irreducible* at the immutable revision *5e35de1*, made public on 3 November 2026.

The two witnesses are $x = P \otimes d_0$, where P is the sum of the six tensors $e_i \otimes e_j \otimes e_k$ with (i,j,k) a permutation of $(0,1,2)$, and the rank-two tensor $y = e_0 \otimes e_0 \otimes e_0 \otimes d_0 + e_1 \otimes e_1 \otimes e_1 \otimes d_1$ (equation (1.3), p. 2). Section 3 (pp. 4–6) proves $\widetilde{R}(P)^3 \le 74 - 3\sqrt[3]{81} < 62$ by a finite Fourier degeneration and Strassen's asymptotic-spectrum and tight-support theorems for three-mode tensors. Corollary 7.2 of Alman and Li (arXiv:2605.21738v1; Corollary 34 of the conference version), cited in Section 1.1, applies to this tensor (their cw'_2) and already gives $\widetilde{R}(P) \le \gamma'_2 \approx 3.9309$ (their Table 1), below $\sqrt[3]{62}$ and below the bound of Section 3. Section 3 gives a self-contained exact derivation; the paper makes no priority claim for its tensor constructions or rank estimates. Section 4 (pp. 6–8) constructs a Zariski-open subset U of all of V containing x on which rank at most three for the three balanced flattenings forces fourth-mode flattening rank at most one (Proposition 4.3, p. 8). With X the closed locus of fourth-mode flattening rank at most one, the proof of Theorem 1.1 at the end of Section 5 (p. 8) gives the cover

$$V_{\le a} = (V_{\le a} \cap X) \cup (V_{\le a} \cap (V \setminus U)),$$

in which y lies outside the first piece and x outside the second.

Comparison with the target. The target asserts irreducibility for every algebraically closed field $\mathbb{F}$, every order $k \ge 3$, all positive integers $d_1, \dots, d_k$ and every real $a \ge 0$, so one reducible sublevel refutes it. Theorem 1.1 is the admissible instance $\mathbb{F} = \mathbb{C}$, $k = 4$, $(d_1, d_2, d_3, d_4) = (3,3,3,2)$ and $a = \sqrt[3]{62}$. The paper's rank is the tensor rank with respect to the four indicated factors, and its asymptotic rank is $\lim_{m\to\infty} R(T^{\boxtimes m})^{1/m}$ with corresponding modes grouped and $\widetilde{R}(0) = 0$ (equation (1.1), p. 1); these are the definitions of the problem statement. The reducible set is the full sublevel in the whole 54-dimensional space, with no linear section and no restriction to a family of tensors. Its two covering pieces are relatively closed and each omits a point of the sublevel, which is exactly the failure of irreducibility described below. The proof of reducibility does not use the closedness theorem of reference 1; that theorem makes $V_{\le a}$ an algebraic variety, which is therefore reducible. The counterexample refutes only the universal assertion. It does not decide the question for order three ($k = 3$), for algebraically

closed fields other than $\mathbb{C}$, or for other formats and thresholds, apart from the further complex cases of order at least four below.

Extensions. For integers $n \geq 3$ and $k \geq 4$, all $d_1, \dots, d_k \geq 2$, and $t_n = \#\{i : d_i \geq n\}$, Theorem 6.5 (p. 12) proves that the complex sublevel in format $(d_1, \dots, d_k)$ has at least $\binom{t_n}{3} + 1$ irreducible components for every real $B_n \leq a < n + 1$, where $n < B_n < n + 1$ is the explicit threshold of Proposition 6.4 (p. 11); the same bound with t_3 holds for $\sqrt[3]{62} \leq a < 4$. The sublevel is therefore reducible whenever $t_n \geq 3$. These are lower bounds, not a classification of components. The counts grow only as the format varies, so they do not settle the fixed-format finiteness question of TR-22. In the other direction, Proposition 6.6 (p. 13) shows that for every $3 \leq a < 4$ the full three-mode sublevel in format $(9, 3, 2)$, obtained from $(3, 3, 3, 2)$ by grouping two three-dimensional modes, is the irreducible variety of 9×6 matrices of rank at most three. This one irreducible three-mode case does not settle the order-three question.

What remains open. Whether every asymptotic-rank sublevel of order three is irreducible, over every algebraically closed field, in every format and at every threshold, remains open. The argument uses three balanced flattenings and the simultaneous slices of a fourth factor, and the grouped example above shows that these constraints do not transfer to three modes (Section 6.5, p. 13). The paper determines neither a complete list of components nor the exact value of $\widetilde{R}(P)$.

Proof and verification scope. Theorem 1.1 and the extensions are proved in the written paper. The cited inputs are Strassen's spectral and tight-support theorems in the formulation of Christandl, Vrana and Zuiddam; the closedness theorem of reference 1, for the interpretation as a variety and in Section 6; and, for the component-growth theorem, a speedup theorem of Alman and Li. A Lean formalization at the same revision proves only a finite certificate layer, recorded in its Lean guide, including the complete enumeration of the appended-minor equations, the rank-26 matrix $L(P)$ with kernel $\mathbb{C}P$, the witness flattening ranks, the finite Fourier exchange with all its Laurent coefficient identities, the tight integer weights and marginal counts, the ten-term identity for the small Coppersmith–Winograd tensor and the integer cutoffs. It does not formalize asymptotic or border rank, the spectral and tight-support theorems, or the passage from entropy to asymptotic subrank. The bound on $\widetilde{R}(P)$, the basis change between the paired and diagonal Coppersmith–Winograd tensors, the balanced-rank-to-kernel equivalence, the ambient open set, the relatively closed cover and every result of Section 6 are proved in the text only. No Lean theorem states the reducibility, so this is not a Lean verification of the target. The verification record also describes exact checks of finite instances, which do not establish the theorem.

Review. Solved by independent informal AI-agent audits, two of the main theorem and two of the component-growth results, together with audits of the Lean statement and of the paper; none requested a mathematical correction. The complete argument was also checked by coauthor Wenqi Zhu (University of Oxford). Qiyuan Chen (Academy of Mathematics and Systems Science, Chinese Academy of Sciences), a coauthor, discussed the proof and agreed with it. The review record contains the audit reports. The paper discloses its use of AI tools. The catalogue did not rerun the Lean or exact checks.

The retained status check below predates this resolution.

Problem statement

Let $\mathbb{F}$ be an algebraically closed field, let $k \geq 3$, and let $d_1, \dots, d_k$ be positive integers. Write $V = \mathbb{F}^{d_1} \otimes \cdots \otimes \mathbb{F}^{d_k}$. For $T \in V$, let $R(T)$ be its tensor rank, and define its asymptotic rank by

$$\widetilde{R}(T) = \lim_{m \to \infty} R(T^{\boxtimes m})^{1/m},$$

where $\boxtimes$ groups corresponding modes of tensor powers; put $\widetilde{R}(0) = 0$.

For every real $a \geq 0$, is the set

$$V_{\leq a} = \{T \in V : \widetilde{R}(T) \leq a\}$$

an irreducible algebraic variety?

The source proves that these sets are Zariski closed. The open assertion is that none can be written as the union of two proper Zariski-closed subsets. All fields, formats, and thresholds above are quantified; $0 \in V_{\leq a}$ avoids the empty-set convention for irreducibility.

Why it matters

For matrices, bounded rank defines an irreducible determinantal variety. The question asks whether a comparable geometric unity survives when tensor rank is replaced by its asymptotic computational cost. A positive answer would constrain the number of costs possible in a format and the ways low-cost tensor families can degenerate.

This is stronger than merely asking for a finite set of asymptotic-rank values. The source lists the two questions separately and records the implication from irreducibility to discreteness; no converse is claimed.

References and status check

1. M. Christandl, K. Hoeberechts, H. Nieuwboer, P. Vrana, and J. Zuiddam, *Asymptotic tensor rank is characterized by polynomials*, arXiv:2411.15789v2, Theorem 1.2 and §5 second bullet, p.17.
2. Their STOC 2025 publication, p.753, "Geometric properties; irreducibility."
3. J. Zuiddam, November 2025 research slides, "Open problems," question (2).

Status check — 2026-09-10

Checked the June 2026 arXiv revision, §5 second bullet, and the STOC 2025 irreducibility discussion, with targeted 2025–2026 searches. The sources leave irreducibility open for tensors of order at least three. Closedness is proved, but neither it nor the excluded matrix case supplies a nontrivial resolved part of this universal irreducibility target. Conditional implications from the full asymptotic-rank conjecture are not unconditional solutions. No general proof or counterexample was located.